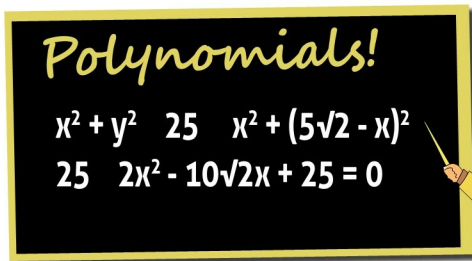


Get Set with Polynomials in Octave

This article on the mathematical journey through open source deals with polynomial mathematics in Octave.



Let's first solve the earlier puzzles before discussing the polynomial power of Octave.

Number puzzle

Find three numbers, the product of which is 60; the sum of their squares is 50; and their sum is 12. Let the X vector elements X(1), X(2) and X(3) be the three numbers. Then, here is the solution:

```
octave:1> function Y = F(X)
> Y(1) = X(1) * X(2) * X(3) - 60;
> Y(2) = X(1)^2 + X(2)^2 + X(3)^2 - 50;
> Y(3) = X(1) + X(2) + X(3) - 12;
> endfunction
octave:2> [Y, Fval, info] = fsolve(@F, [3; 3; 3])
warning: matrix singular to machine precision, rcond =
4.32582e-35
warning: attempting to find minimum norm solution
warning: dgelsd: rank deficient 3x3 matrix, rank = 1
Y =
```

```
5.0000
3.0000
4.0000
```

Fval =

```
-3.2345e-07 1.0351e-07 0.0000e+00
```

info = 1

```
octave:3>
```

So, the three numbers are 5, 3 and 4.

Flower puzzle

A sage came to a temple with some flowers and dipped them into the first pond of the temple to get them squared. Then, he offered some flowers to the idol and dipped the remaining flowers into the second pond to get them doubled. Later, he again offered the idol the same number of flowers, as earlier, and dipped the remaining flowers into the third pond to get them tripled to take back with him as *prasadam* (this was the same number as in each one of his earlier offerings). Now, if he took back thrice the number of flowers he brought, how many did he bring in with him?

Let the X vector elements X(1) and X(2) be the number of flowers the sage came with, and the number of flowers the sage offered each time, respectively. Here is the solution:

```
octave:1> function y = f(x)
> y(1) = ((x(1) * x(1) - x(2)) * 2 - x(2)) * 3 - x(2);
> y(2) = x(2) - 3 * x(1);
> endfunction
octave:2> [x fval info] = fsolve(@f, [10; 10])
x =
```

```
5.0000
15.0000
```

fval =

```
-2.8791e-06 -1.7764e-15
```